

Worked example: IAP × RPG interaction

Outcome 1 — Install success | Pooled analysis

1 Construct the two exposure-by-outcome tables

RPG stratum (n = 133,476)

	IAP = 1	IAP = 0	Total
Success	62,740	8,855	71,595
Not success	42,174	19,707	61,881
Total	104,914	28,562	133,476

Non-RPG stratum (n = 1,645,224)

	IAP = 1	IAP = 0	Total
Success	361,497	456,562	818,059
Not success	455,405	371,760	827,165
Total	816,902	828,322	1,645,224

2 Estimate conditional success probabilities

RPG

Outcome	IAP = 1	IAP = 0
Success	62,740 / 104,914 = 0.5980	8,855 / 28,562 = 0.3100

Non-RPG

Outcome	IAP = 1	IAP = 0
Success	361,497 / 816,902 = 0.4425	456,562 / 828,322 = 0.5512

$$\Delta p_{\text{RPG}} = 0.5980 - 0.3100 = +0.2880$$

$$OR_{\text{RPG}} = \frac{62740 \times 19707}{42174 \times 8855} = 3.3108$$

$$\Delta p_{\text{Non-RPG}} = 0.4425 - 0.5512 = -0.1087$$

$$OR_{\text{Non-RPG}} = \frac{361497 \times 371760}{455405 \times 456562} = 0.6464$$

Pooled interaction analysis

Install success | Additive and multiplicative scales

3 Additive interaction

$$DDP = \Delta p_{RPG} - \Delta p_{Non-RPG}$$

$$= +0.2880 - (-0.1087) = +0.3967$$

95% CI: [0.3903, 0.4030]

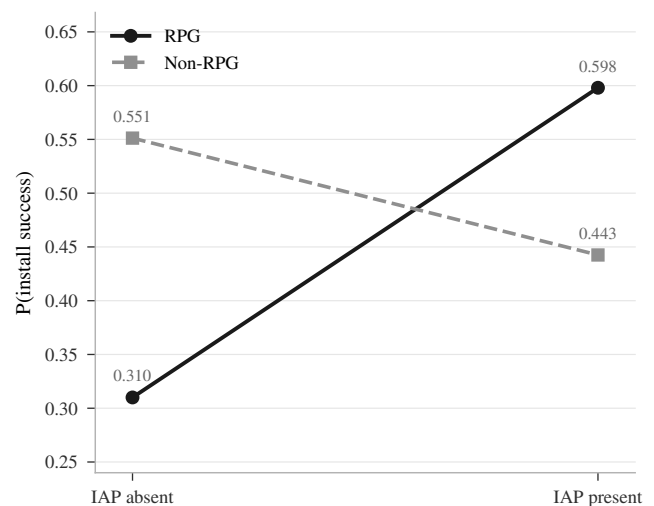
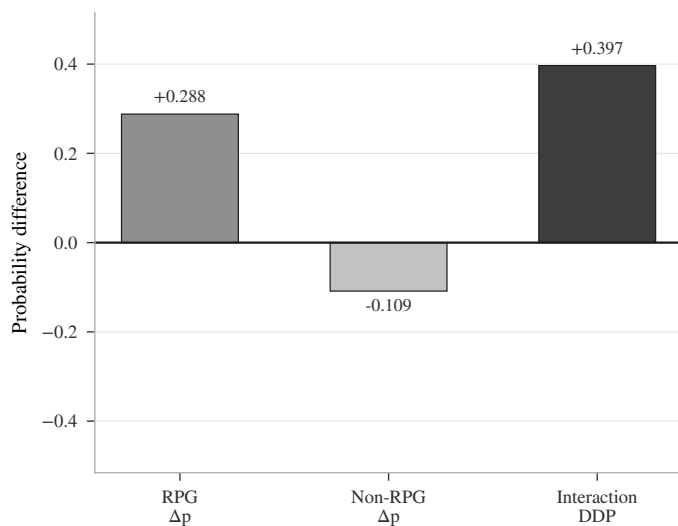


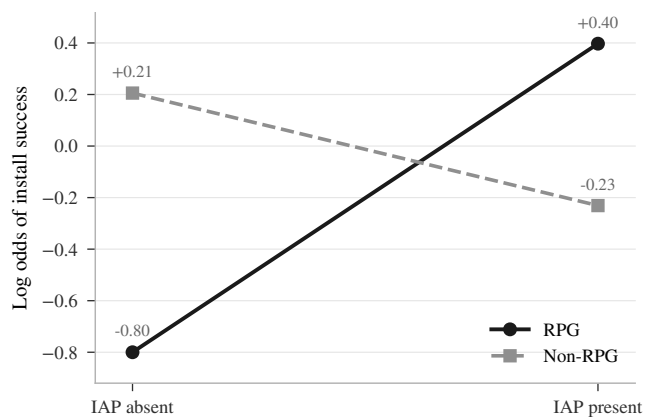
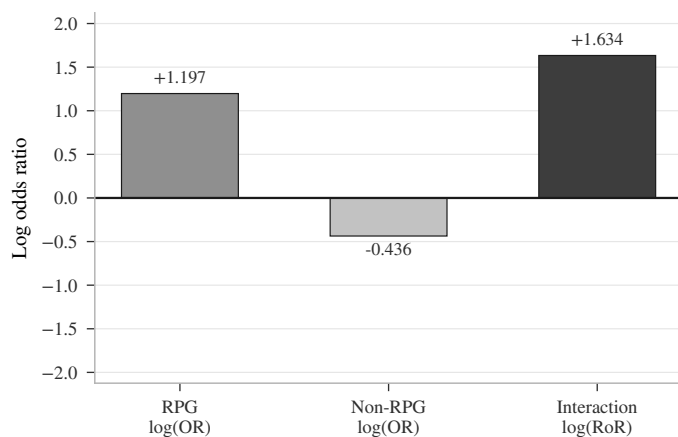
Figure 1. Additive interaction. The left panel decomposes the two within-stratum probability differences and their difference-in-differences; the right panel shows the corresponding conditional success probabilities.

4 Multiplicative interaction

$$RoR = OR_{RPG} / OR_{Non-RPG}$$

$$= 3.3108 / 0.6464 = 5.1223$$

95% CI: [4.9778, 5.2710]



Cross-market replication

Install success | United States and India

Country-specific calculations

United States

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
RPG	1,916	838	150	274
Non-RPG	10,287	12,696	8,829	7,360

$$\Delta p_{\text{RPG}} = 0.6957 - 0.3538 = +0.3419$$

$$\Delta p_{\text{Non-RPG}} = 0.4476 - 0.5454 = -0.0978$$

$$\text{DDP} = +0.3419 - (-0.0978) = \mathbf{+0.4397}$$

$$\text{RoR} = 4.1765/0.6754 = \mathbf{6.1833}$$

Joint: ++

India

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
RPG	593	1,240	347	998
Non-RPG	4,973	6,853	15,269	12,077

$$\Delta p_{\text{RPG}} = 0.3235 - 0.2580 = +0.0655$$

$$\Delta p_{\text{Non-RPG}} = 0.4205 - 0.5584 = -0.1378$$

$$\text{DDP} = +0.0655 - (-0.1378) = \mathbf{+0.2034}$$

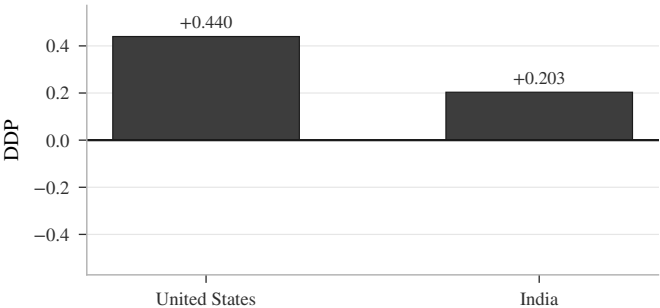
$$\text{RoR} = 1.3754/0.5740 = \mathbf{2.3963}$$

Joint: ++

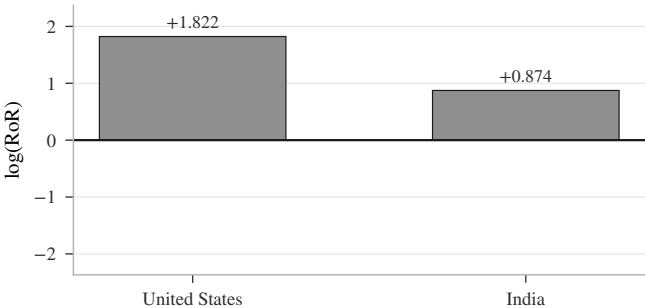
Comparison of interaction estimates

Context	RPG Δp	Non-RPG Δp	DDP	RPG OR	Non-RPG OR	RoR	Joint
United States	+0.342	-0.098	+0.440	4.176	0.675	6.183	++
India	+0.066	-0.138	+0.203	1.375	0.574	2.396	++

Additive interaction



Multiplicative interaction



Temporal replication

Install success | Three United States monthly contexts

Selected monthly contexts

Jan 2025

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
RPG	142	61	5	19
Non-RPG	731	948	635	484

$$\Delta p_{\text{RPG}} = 0.6995 - 0.2083 = +0.4912$$

$$\Delta p_{\text{Non-RPG}} = 0.4354 - 0.5675 = -0.1321$$

$$\text{DDP} = +0.4912 - (-0.1321) = \mathbf{+0.6233}$$

$$\text{RoR} = 8.8459/0.5877 = \mathbf{15.0509}$$

Joint: ++

Aug 2025

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
RPG	128	66	9	24
Non-RPG	720	907	656	515

$$\Delta p_{\text{RPG}} = 0.6598 - 0.2727 = +0.3871$$

$$\Delta p_{\text{Non-RPG}} = 0.4425 - 0.5602 = -0.1177$$

$$\text{DDP} = +0.3871 - (-0.1177) = \mathbf{+0.5047}$$

$$\text{RoR} = 5.1717/0.6232 = \mathbf{8.2986}$$

Joint: ++

Feb 2026

Stratum	IAP+ Success	IAP+ No	IAP- Success	IAP- No
RPG	123	65	12	27
Non-RPG	757	877	621	543

$$\Delta p_{\text{RPG}} = 0.6543 - 0.3077 = +0.3466$$

$$\Delta p_{\text{Non-RPG}} = 0.4633 - 0.5335 = -0.0702$$

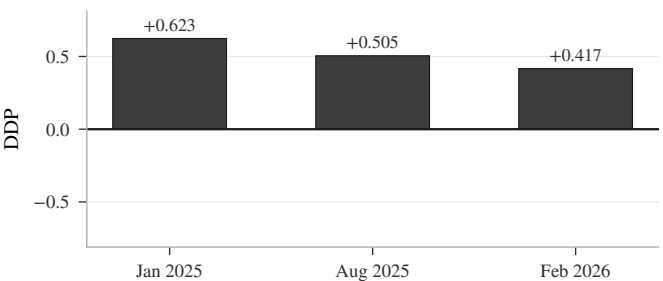
$$\text{DDP} = +0.3466 - (-0.0702) = \mathbf{+0.4168}$$

$$\text{RoR} = 4.2577/0.7548 = \mathbf{5.6412}$$

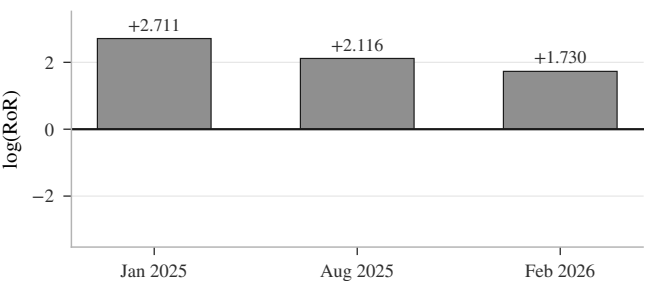
Joint: ++

Temporal comparison

Additive interaction



Multiplicative interaction



Install-success synthesis

Pooled, national, and temporal contexts

Interaction estimates

Context	RPG Δp	Non-RPG Δp	DDP	RPG OR	Non-RPG OR	RoR	Joint
Pooled	+0.288	-0.109	+0.397	3.311	0.646	5.122	++
United States	+0.342	-0.098	+0.440	4.176	0.675	6.183	++
India	+0.066	-0.138	+0.203	1.375	0.574	2.396	++
US Jan 2025	+0.491	-0.132	+0.623	8.846	0.588	15.051	++
US Aug 2025	+0.387	-0.118	+0.505	5.172	0.623	8.299	++
US Feb 2026	+0.347	-0.070	+0.417	4.258	0.755	5.641	++

Interaction magnitude across contexts

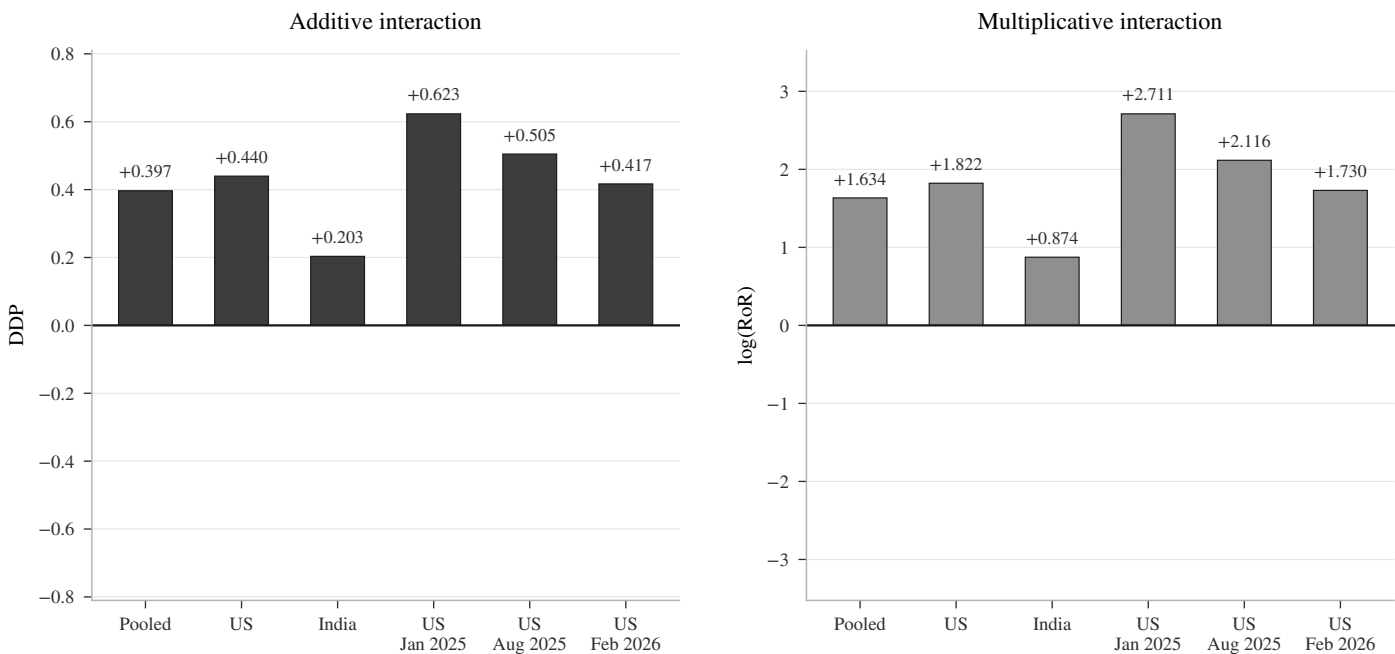


Figure 2. Install-success interaction estimates across the pooled analysis, two national markets, and three United States monthly contexts. Both panels use a zero null: DDP on the additive scale and log(RoR) on the multiplicative scale.

The worked contexts are presented to demonstrate how the same interaction calculation is propagated across analytical levels. The table retains RoR on its original ratio scale for reporting, whereas the figure uses log(RoR) so that the multiplicative null is centered at zero.

Parallel outcome analysis

Outcome 2 — Rating success | Pooled analysis

1 Construct the two exposure-by-outcome tables

RPG				Non-RPG			
	IAP = 1	IAP = 0	Total		IAP = 1	IAP = 0	Total
Success	63,345	12,726	76,071	Success	402,800	422,194	824,994
Not success	41,569	15,836	57,405	Not success	414,102	406,128	820,230
Total	104,914	28,562	133,476	Total	816,902	828,322	1,645,224

2 Estimate within-stratum associations

Stratum	P(success IAP = 1)	P(success IAP = 0)	Δp	OR
RPG	0.6038	0.4456	+0.1582	1.8963
Non-RPG	0.4931	0.5097	-0.0166	0.9357

3 Estimate interaction on both scales

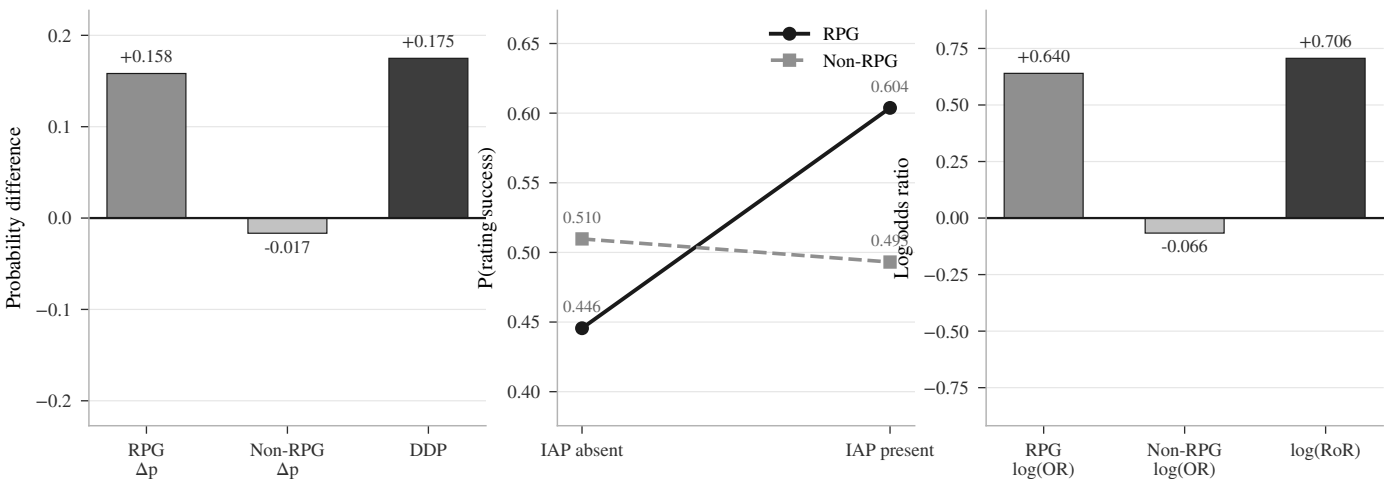
$$\text{DDP} = +0.1582 - (-0.0166) = \mathbf{+0.1748}$$

$$\text{RoR} = 1.8963 / 0.9357 = \mathbf{2.0266}$$

95% CI: [0.1682, 0.1815]

95% CI: [1.9724, 2.0823]

Joint classification: ++



Rating-success replication

One national context and one monthly context

Context-specific calculations

United States

Stratum	IAP+ Success	IAP+ No	IAP– Success	IAP– No
RPG	1,723	1,031	180	244
Non-RPG	11,427	11,556	8,157	8,032

$$\Delta p_{\text{RPG}} = 0.6256 - 0.4245 = +0.2011$$

$$\Delta p_{\text{Non-RPG}} = 0.4972 - 0.5039 = -0.0067$$

$$\text{DDP} = +0.2011 - (-0.0067) = \mathbf{+0.2078}$$

$$\text{RoR} = 2.2654/0.9737 = \mathbf{2.3266}$$

Joint: ++

United States — Aug 2025

Stratum	IAP+ Success	IAP+ No	IAP– Success	IAP– No
RPG	127	67	16	17
Non-RPG	812	815	572	599

$$\Delta p_{\text{RPG}} = 0.6546 - 0.4848 = +0.1698$$

$$\Delta p_{\text{Non-RPG}} = 0.4991 - 0.4885 = +0.0106$$

$$\text{DDP} = +0.1698 - (+0.0106) = \mathbf{+0.1592}$$

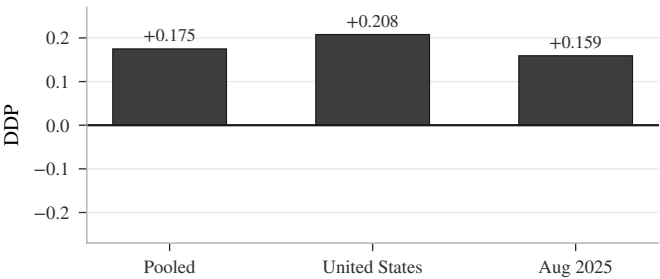
$$\text{RoR} = 2.0140/1.0433 = \mathbf{1.9303}$$

Joint: 00

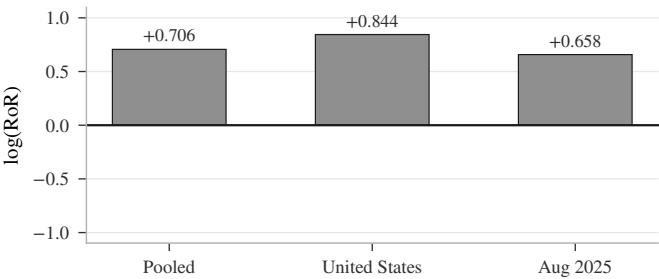
Rating-success summary

Context	RPG Δp	Non-RPG Δp	DDP	RPG OR	Non-RPG OR	RoR	Joint
Pooled	+0.158	-0.017	+0.175	1.896	0.936	2.027	++
United States	+0.201	-0.007	+0.208	2.265	0.974	2.327	++
US Aug 2025	+0.170	+0.011	+0.159	2.014	1.043	1.930	00

Additive interaction



Multiplicative interaction



Complete IAP × RPG interaction summary

Pooled, cross-market, and temporal evidence

Interaction estimates across analytical levels

Outcome	Analytical level	Mean / pooled DDP	95% CI / observed range	Mean / pooled RoR	Mean / pooled log(RoR)	++	--	00	Mixed	N
Install	Pooled	+0.397	[+0.390, +0.403]	5.122	+1.634	1	0	0	0	1
Install	Countries	+0.367	+0.091 to +0.538	5.030	+1.524	42	0	0	0	42
Install	Country × month	+0.368	-0.009 to +0.725	5.648	+1.552	551	0	29	8	588
Rating	Pooled	+0.175	[+0.168, +0.181]	2.027	+0.706	1	0	0	0	1
Rating	Countries	+0.163	+0.095 to +0.279	1.961	+0.661	42	0	0	0	42
Rating	Country × month	+0.163	-0.113 to +0.514	2.133	+0.671	257	0	325	6	588

Pooled rows report the interaction estimate and its 95% confidence interval. Country and country × month rows summarize the complete set of context-specific estimates; the interval column therefore shows the observed range rather than a confidence interval.

Cross-market classification

Outcome	Mean DDP	Mean RoR	++	--	00	Mixed	Countries
Install success	+0.367	5.030	42	0	0	0	42
Rating success	+0.163	1.961	42	0	0	0	42

Country × month classification

Outcome	Mean DDP	Mean RoR	++	--	00	Mixed	Contexts
Install success	+0.368	5.648	551	0	29	8	588
Rating success	+0.163	2.133	257	0	325	6	588

Joint classification combines additive and multiplicative inference: ++ indicates statistically positive interaction on both scales; -- indicates statistically negative interaction on both scales; 00 indicates null interaction on both scales; Mixed contains all remaining combinations. Multiplicative effects are reported as RoR in the tables and transformed to log(RoR) when a zero-centered visual scale is required.

Country-specific interaction estimates

IAP × RPG interaction | All national markets

Install and rating interaction estimates by country

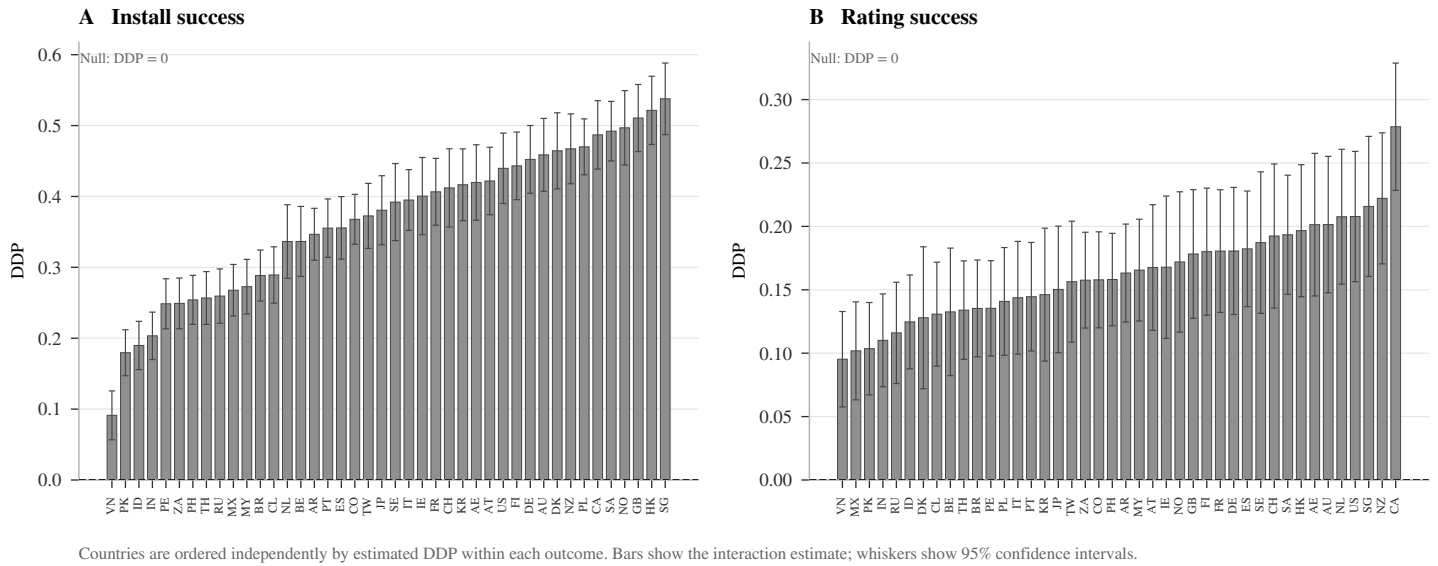
Country	Install DDP [95% CI]	Install RoR [95% CI]	Rating DDP [95% CI]	Rating RoR [95% CI]
AE	+0.420 [+0.366, +0.473]	5.632 [4.411, 7.190]	+0.201 [+0.145, +0.258]	2.258 [1.792, 2.845]
AR	+0.347 [+0.310, +0.383]	4.231 [3.583, 4.995]	+0.163 [+0.125, +0.202]	1.929 [1.649, 2.258]
AT	+0.422 [+0.374, +0.469]	5.690 [4.617, 7.012]	+0.168 [+0.118, +0.217]	1.986 [1.625, 2.428]
AU	+0.459 [+0.407, +0.510]	6.690 [5.319, 8.415]	+0.201 [+0.148, +0.255]	2.267 [1.821, 2.822]
BE	+0.337 [+0.287, +0.386]	3.910 [3.182, 4.805]	+0.133 [+0.082, +0.183]	1.708 [1.395, 2.091]
BR	+0.288 [+0.252, +0.324]	3.383 [2.869, 3.989]	+0.135 [+0.097, +0.173]	1.722 [1.476, 2.010]
CA	+0.487 [+0.439, +0.535]	7.761 [6.212, 9.697]	+0.279 [+0.228, +0.329]	3.127 [2.525, 3.872]
CH	+0.412 [+0.357, +0.467]	5.461 [4.309, 6.920]	+0.192 [+0.136, +0.249]	2.196 [1.745, 2.764]
CL	+0.289 [+0.250, +0.329]	3.255 [2.743, 3.863]	+0.131 [+0.090, +0.172]	1.693 [1.434, 1.998]
CO	+0.368 [+0.333, +0.403]	4.803 [4.060, 5.681]	+0.158 [+0.120, +0.196]	1.889 [1.618, 2.205]
DE	+0.452 [+0.404, +0.500]	6.524 [5.273, 8.073]	+0.181 [+0.130, +0.231]	2.090 [1.706, 2.561]
DK	+0.464 [+0.411, +0.518]	7.021 [5.547, 8.888]	+0.128 [+0.072, +0.184]	1.689 [1.348, 2.116]
ES	+0.356 [+0.312, +0.400]	4.262 [3.517, 5.164]	+0.182 [+0.137, +0.228]	2.086 [1.732, 2.511]
FI	+0.443 [+0.396, +0.491]	6.212 [5.016, 7.694]	+0.180 [+0.130, +0.230]	2.077 [1.694, 2.546]
FR	+0.407 [+0.359, +0.454]	5.333 [4.360, 6.524]	+0.181 [+0.132, +0.229]	2.086 [1.714, 2.539]
GB	+0.511 [+0.463, +0.558]	8.582 [6.886, 10.697]	+0.178 [+0.128, +0.229]	2.076 [1.690, 2.549]
HK	+0.521 [+0.473, +0.570]	9.132 [7.262, 11.482]	+0.197 [+0.145, +0.249]	2.231 [1.805, 2.757]
ID	+0.190 [+0.156, +0.224]	2.237 [1.900, 2.634]	+0.125 [+0.088, +0.162]	1.651 [1.421, 1.918]
IE	+0.401 [+0.346, +0.455]	5.216 [4.130, 6.587]	+0.168 [+0.112, +0.224]	1.989 [1.586, 2.495]
IN	+0.203 [+0.170, +0.237]	2.396 [2.037, 2.819]	+0.110 [+0.074, +0.147]	1.557 [1.343, 1.805]
IT	+0.395 [+0.352, +0.438]	5.024 [4.169, 6.055]	+0.144 [+0.099, +0.188]	1.792 [1.497, 2.145]
JP	+0.381 [+0.332, +0.429]	4.775 [3.885, 5.869]	+0.150 [+0.100, +0.200]	1.844 [1.507, 2.255]
KR	+0.417 [+0.366, +0.467]	5.582 [4.484, 6.949]	+0.146 [+0.094, +0.199]	1.820 [1.472, 2.249]
MX	+0.268 [+0.231, +0.304]	3.087 [2.616, 3.643]	+0.102 [+0.063, +0.140]	1.504 [1.288, 1.757]
MY	+0.273 [+0.234, +0.311]	3.102 [2.613, 3.682]	+0.166 [+0.125, +0.206]	1.947 [1.652, 2.295]
NL	+0.336 [+0.285, +0.388]	3.926 [3.137, 4.913]	+0.208 [+0.154, +0.261]	2.313 [1.858, 2.879]
NO	+0.497 [+0.444, +0.549]	8.070 [6.371, 10.223]	+0.172 [+0.117, +0.227]	2.032 [1.624, 2.542]
NZ	+0.467 [+0.418, +0.516]	7.023 [5.626, 8.767]	+0.222 [+0.170, +0.274]	2.483 [2.010, 3.067]
PE	+0.249 [+0.213, +0.284]	2.882 [2.450, 3.391]	+0.135 [+0.098, +0.173]	1.723 [1.480, 2.006]
PH	+0.254 [+0.220, +0.289]	2.937 [2.510, 3.437]	+0.158 [+0.122, +0.195]	1.889 [1.628, 2.192]
PK	+0.180 [+0.147, +0.212]	2.086 [1.765, 2.465]	+0.104 [+0.067, +0.140]	1.520 [1.310, 1.763]
PL	+0.470 [+0.431, +0.509]	7.117 [5.916, 8.563]	+0.141 [+0.098, +0.183]	1.772 [1.492, 2.104]
PT	+0.355 [+0.314, +0.397]	4.262 [3.561, 5.101]	+0.145 [+0.102, +0.187]	1.796 [1.510, 2.135]
RU	+0.259 [+0.221, +0.298]	2.922 [2.469, 3.456]	+0.116 [+0.076, +0.156]	1.593 [1.357, 1.871]
SA	+0.492 [+0.450, +0.534]	8.059 [6.498, 9.993]	+0.193 [+0.146, +0.240]	2.197 [1.813, 2.661]
SE	+0.392 [+0.338, +0.446]	4.969 [3.936, 6.274]	+0.187 [+0.131, +0.243]	2.134 [1.701, 2.679]
SG	+0.538 [+0.487, +0.588]	9.778 [7.633, 12.527]	+0.216 [+0.161, +0.271]	2.410 [1.923, 3.021]
TH	+0.257 [+0.220, +0.294]	2.903 [2.463, 3.423]	+0.134 [+0.095, +0.173]	1.713 [1.464, 2.005]
TW	+0.373 [+0.327, +0.418]	4.579 [3.746, 5.597]	+0.156 [+0.109, +0.204]	1.881 [1.551, 2.282]
US	+0.440 [+0.390, +0.489]	6.183 [4.968, 7.695]	+0.208 [+0.156, +0.259]	2.327 [1.883, 2.874]
VN	+0.091 [+0.057, +0.126]	1.372 [1.162, 1.620]	+0.095 [+0.058, +0.133]	1.465 [1.259, 1.706]
ZA	+0.249 [+0.213, +0.285]	2.878 [2.440, 3.395]	+0.158 [+0.120, +0.195]	1.890 [1.618, 2.208]

Each row reports the country-specific IAP × RPG interaction estimate and its 95% confidence interval. DDP is the additive interaction (null = 0); RoR is the multiplicative interaction (null = 1). The table reports the complete national-market estimates rather than cross-country means.

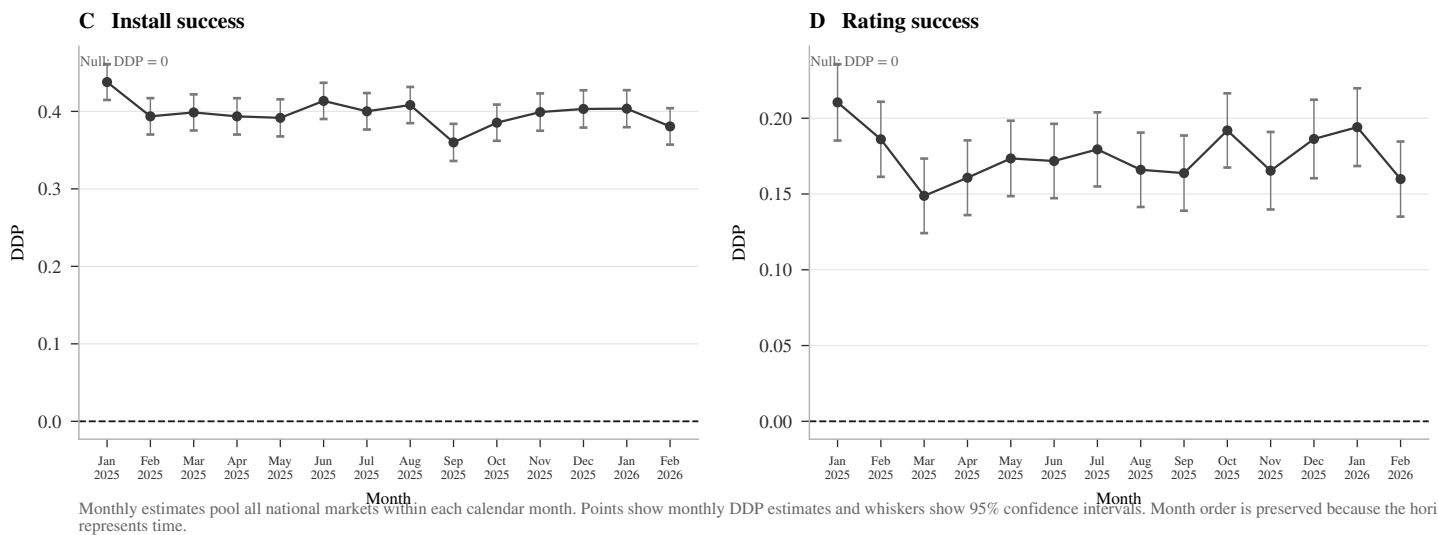
Complete cross-market and temporal synthesis

IAP $\times$ RPG interaction | Install success and rating success

Cross-market interaction — all national markets



Temporal interaction — all 14 pooled months



Reading the figure

DDP = 0 is the additive no-interaction reference. Values below zero indicate that the IAP-associated probability difference is more negative for RPG than for non-RPG games; values above zero indicate the reverse. The corresponding multiplicative evidence is retained in the complete numerical synthesis on the preceding page using RoR and log(RoR).